Exploring a connection between air quality and labour productivity

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Abstract

This paper explores the correlation between air quality, specifically PM2.5 concentration, and labour productivity across 138 countries and territories. Using publicly available data from IQAir World Air Quality Ranking [3] and International Labour Organization [21] for the years 2022, 2023 and 2024, the analysis is aimed to determine whether air quality is associated with labour productivity. A moderate negative correlation was found, averaging about 0.5, suggesting that lower PM2.5 levels are linked to greater economic output per worker.

Jel classification

Q53 - Air Pollution; Water Pollution; Noise; Hazardous Waste; Solid Waste; Recycling

O47 - Measurement of Economic Growth; Aggregate Productivity; Cross-Country Output Convergence

J24 - Human Capital; Skills; Occupational Choice; Labor Productivity

Keywords

PM2.5, Labour Productivity, Air Pollution, Global Productivity

Introduction

During the Industrial Revolution, factories were the main driver of economic prosperity. They meant jobs and wealth, and dirty air was a normal part of progress. But today, many countries make more money through services and specialized work that relies more on brainpower than machines, while heavy industry has often moved abroad in an effort to minimize costs. Air pollution now looks less like a sign of progress and more like a drag on it. Breathing polluted air every day can cause health problems like headaches, fatigue, and breathing issues, and it can even reduce how well people think and concentrate. That means workers may get less done, make more mistakes, or need more sick days, which would make it seem that regardless of what is the source wealth in a country, having cleaner air should improve the productivity of its workers. On top of that, skilled workers often prefer to live in clean, healthy environments, so places with better air not only help people work better but also attract the kind of people who drive modern economies. This study will thus look at the link between air pollution and labour productivity The main hypothesis is whether is air pollution correlated with labour productivity on a global scale.

Theoretical part

Air quality index

There is no single way of measuring air pollution across the world. Most of the time it is measured by an air quality index (AQI). Generally, the index combines measurements of concentrations of pollutant chemical compounds, (such as SO_2, NO_2, CO, O_3) and a concentration of particulate matter smaller than 10 μm (PM_{10}) and 2.5 μm ($PM_{2.5}$). Many countries use their own version of an air quality index to measure the pollution, assigning a varying level of importance to each compound measured. One such example is European Air Quality Index (EAQI) [1] used for measuring air quality across Europe. An approach that worldwide indexes often use is to only look at the $PM_{2.5}$ concentrations, which are often the dominant pollutant and according to a study from 2020 [2] strongly correlate with other pollutants. Pollution will thus be measured as an **annual average $PM_{2.5}$ concentration (μg/m^3).** As an example, the most polluted country in 2024 was Chad with 91.8 μg/m^3 ,and least were Bahamas with 2.3 μg/m^3 (most polluted city was Byrnihat, India with 128.2 μg/m^3) [3]. WHO advises that ideally concentrations should not exceed 5 μg/m^3 [1].

Causes of air pollution

Air pollution at its core reflects a modern lifestyle. Most of the human causes of air pollution are not strictly necessary for our survival but merely serve to enhance our convenience and comfort. It would make sense that a country's air quality would be linked to its productive output. Industrial activity remains a major contributor, a study in China calculated that manufacturing and construction accounted for over 50% of air pollution in 2014 [4]. In the UK, overall $PM_{2.5}$ emissions have dropped by 76% since 1990, while industries still release tens of thousands of tonnes annually, the number of released particles has dropped across the board. Industrial combustion dropped from 28.8 thousand tonnes in 1990 to 5.8 in 2023, and industrial processes fell from 101.2 to 42.7. The major causes of pollutants tied to the individual have also decreased. Domestic combustion, mainly from heating and stoves, fell from 42.1 to 11.5 and

road transport from 37.6 to 20.1 thousand tonnes in the same period [5]. The areas responsible for the most pollution are often those most heavily relied upon by developing countries. Rich countries moving away from heavy industry experience a decreasing rate of air pollution.

Another source of pollution is electricity production, especially from lignite, coal, and oil [6]. According to more studies, agriculture contributes significantly through fertilizer use to nitrogen oxides [7] and livestock to ground level ozone [8]. A WHO report mentions two sources disproportionately impacting low-income countries. Open waste dumps and household cooking, which by itself is responsible for 12% of $PM_{2.5}$ particles globally [9]. Particularly polluting is cooking on firewood or coal highlighting that some of the pollution can be directly linked to the development of the country.

An analysis of natural sources of $PM_{2.5}$, such as wildfires, dust storms, sea salt, and pollen in 2008 and 2009 under the EU Air Quality Directive [10] found that they can contribute significantly in certain regions or seasons but are generally less consistent and widespread than human made emissions.

Health risks of air pollution

A direct undisputable effect on labour productivity are the health complications that come with increased air pollution. The most visible outcome is increased mortality. A State of Global Air Report 2024 [11] claims that in 2021 air pollution accounted for 8.1 million deaths, which in total is more than 1 in 8. Diseases including heart disease, stroke, diabetes, lung cancer, and chronic obstructive pulmonary disease (COPD) account for nearly 90% of the disease burden from air pollution. The largest number of deaths from pollution was attributed to particulate pollution ($PM_{2.5}$). They estimated that air pollution exposure shortens the average person's lifespan by 1.8 years. One could also assume that people in places with better air quality live an overall healthier life and experience less day to day health complications. While this is hard to measure, there have been several studies showing a positive correlation between rate of pollution and sick leave. For example, Beshir and Fichera (2025) [12] found that an introduction of low emission and ultra-low emission zones in London reduced sick leave by 18.5%. Another assumption would be that clear air allows us to think more clearly. It has been long known that indoor air quality, particularly CO_2 affects cognitive performance, and while the effect is not as pronounced with outdoor air, it is still significant. Especially with attention driven tasks. Künn, Palacios, and Pestel (2023) [13] conducted a study of chess players found that for every $10\mu m/m^3$ increase of $PM_{2.5}$ chess players are 2.1% more likely to make a significant error and they are 20.2% more likely to commit mistakes throughout the match. Such mistakes can easily translate into a loss of productivity.

Desire for cleaner air

Striving for cleaner air is a common theme across all continents as it provides many other benefits not necessarily connected to increasing labour productivity. Most countries have implemented at least some measures to improve air quality. In public opinion these are often thought of as a luxury which comes at a monetary cost. On average, it could be assumed that richer countries are able to afford more effective air quality policies. However, a reversed effect is likely to also be at play. As two Chinese studies show, one tracking migration within the

country [14] and one focusing on international migration [15], people are willing to move to places with better air quality. Within China, a 10 percent increase in air pollution, holding everything else constant, is capable of reducing population through net outmigration by about 2.8 percent in a given county. This inherently makes cleaner places a more attractive destination for skilled work and capital.

Previous studies

The link between air pollution and labour productivity has been extensively studied in the past. Many of the studies focus on a particular company, sector or a country and attempt to compare workers performance relative to changing levels of air pollution. Chang, Zivin, Gross, and Neidell (2019) examine the relationship between air pollution and the productivity of individual workers at a large call center in China. “We find that a 10-unit increase in the air pollution index decreases the number of daily calls handled by a worker by 0.35 percent. Our analysis also suggests that these productivity losses are largely linearly increasing in pollution levels.” [16]

Countries with clean air also seem to benefit from further improvements. Kögel, C. (2022) found that for French workers a 10% increase in fine particulate matter leads, on average, to a 1.5% decrease in labour productivity, controlling for firm-specific characteristics and other confounding factors. [17].

Chen, S., & Zhang, D. (2021) studied a prison factory in Shenzen, and observed that a 10-unit increase in air pollution index led to a significant decrease in labour productivity by 4%. [18]

There are a few studies which come closer to the examined hypothesis and attempt to check if air pollution is correlated to the whole economy of a state. A study of regions across European Union by Dechezleprêtre, A., N. Rivers and B. Stadler (2019) [19] estimates that a 1μg/m3 increase in PM2.5 concentration (or a 10% increase at the sample mean) causes a 0.8% reduction in real GDP that same year. Ninety-five per cent of this impact is due to reductions in output per worker, which can occur through greater absenteeism at work or reduced labour productivity.

And finally, Dong, D., & Wang, J. (2023). An analysis of agricultural data from 146 countries estimates, that a 1% increase in $PM_{2.5}$ and O_3 concentration would cause the agricultural total factor productivity (TFP) to decline by 0.104% and 0.207%, respectively.

Based on previous results there exists a good chance a correlation between air pollution and labour productivity will be found even at a global scale.

Labour productivity has a few slightly different definitions. For the purposes of this paper, it will be calculated as **GDP per hour worked (Int$ PPP).**

Practical part

Air quality data was taken from the website iqair.com from the IQAir World Air Quality Ranking [3]. It provides data for 138 places, most of which are coutnries and a few are autonomous territories. This will represent the sample size.

Data for labour productivity was taken from the International Labour Organization (ILO) [21], which provides data on all the selected places.

The countries cover most of the range for both air quality and labour productivity, as well as geographic location and population sizes. None of the places seem to be systematically chosen or excluded. From the countries missing most are either microstates, small island nations or have a very low level of infrastructure and likely do not capture air quality data at scale. These places also tend to be the ones with an unstable political situation or a very small area, which could potentially skew results. Nonetheless for the sake of completeness unstable countries and microstates already in the dataset were not manually removed.

The first step was to identify the outliers in the data. Chad, Bangladesh, Pakistan, Democratic Republic of the Congo, India, Tajikistan, Kuwait, Bahrain and Iraq had too high levels of pollution. While Luxembourg, Norway and Ireland had an excessive rate of labour productivity.

The statistical analysis will test the **null hypothesis, that air quality and labour productivity are not correlated on global scale**, using a confidence level of 95% and an acceptable margin of error of 7%. Assuming 195 countries or 241 countries and territories, 98 and 109 samples would be needed respectively. Both of which the dataset exceeds.

The analyzed data was from the years 2022, 2023 and 2024 using simple linear regression. All the graphs show a negative correlation between air pollution and labour productivity, usually around 0.5. Tables 1,2 and 3 then show the Pearson correlation coefficient, R Squared, Intercept, Slope, p-value and standard error. The p-value consistently falls well below 0.05, rejecting the null hypothesis on a confidence level of 95%.

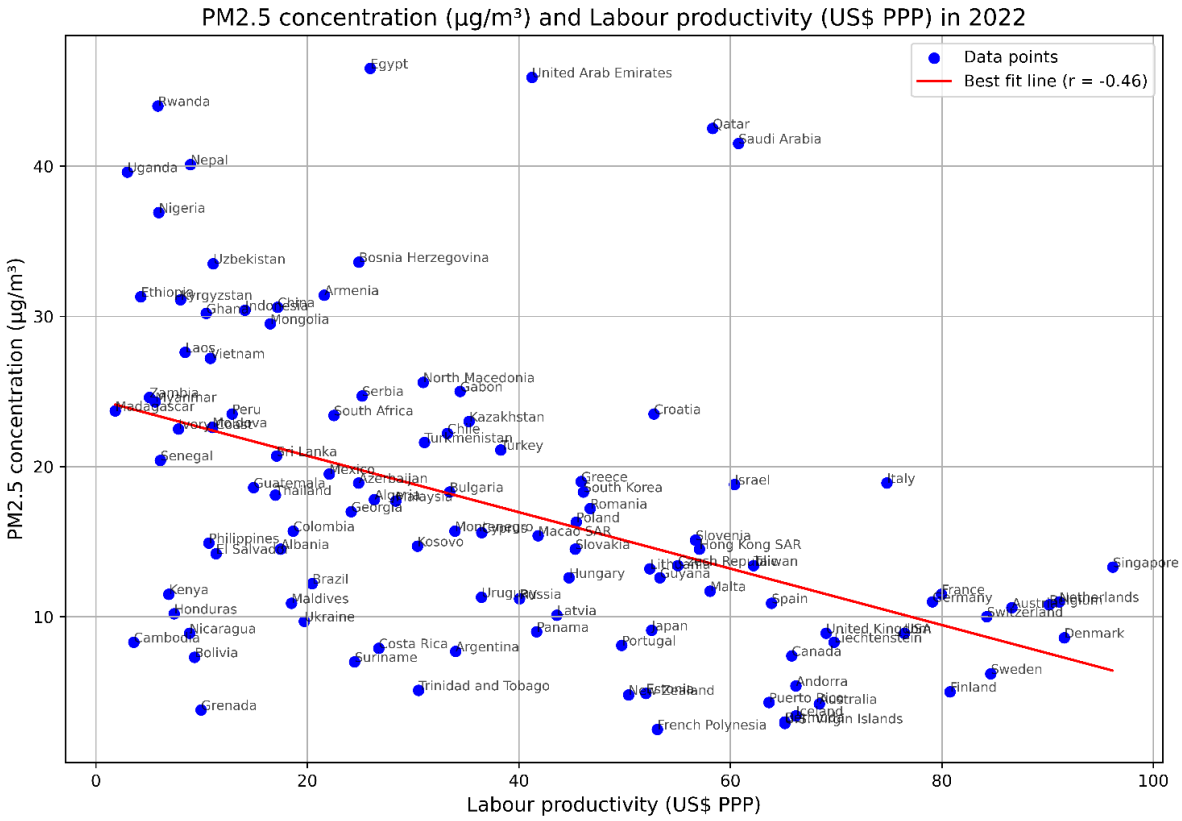


Figure 1: PM2.5 concentration (µg/m^3) and Labour productivity (US$ PPP) in 2022, data sources: IQAir [3] and ILO [21]

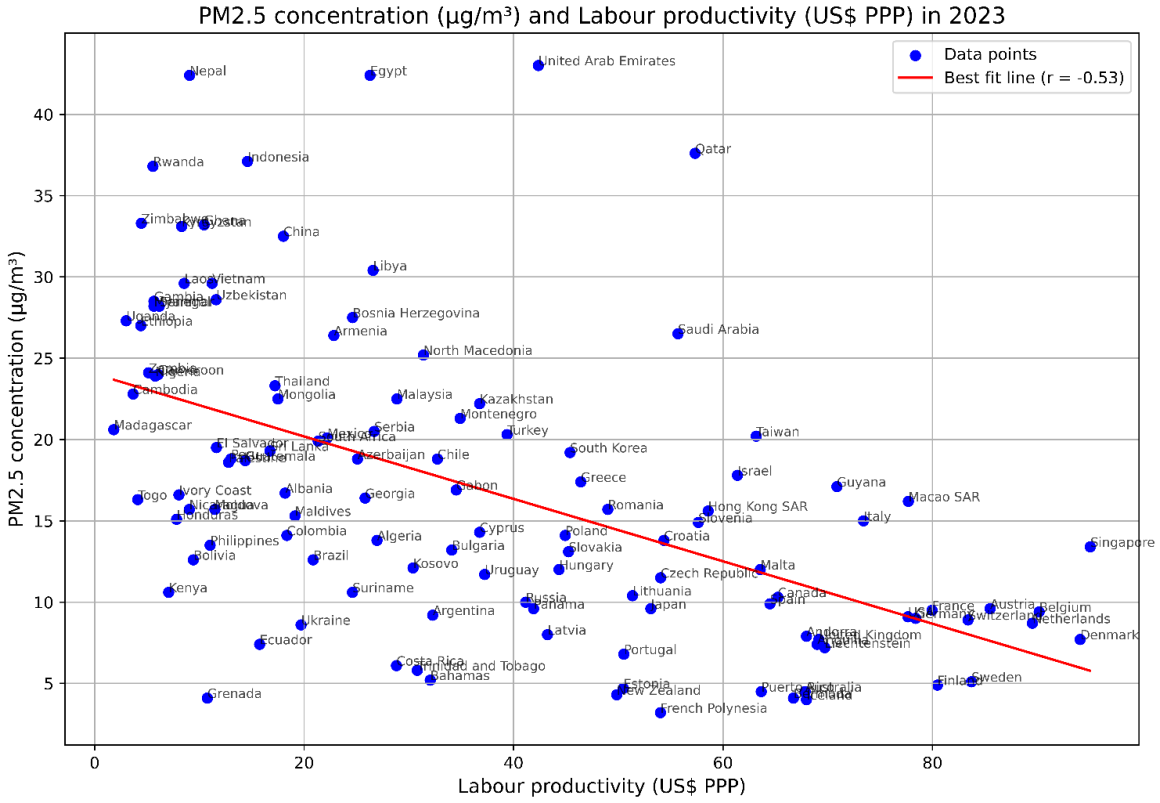


Figure 2: PM2.5 concentration (µg/m^3) and Labour productivity (US$ PPP) in 2023, data sources: IQAir [3] and ILO [21]

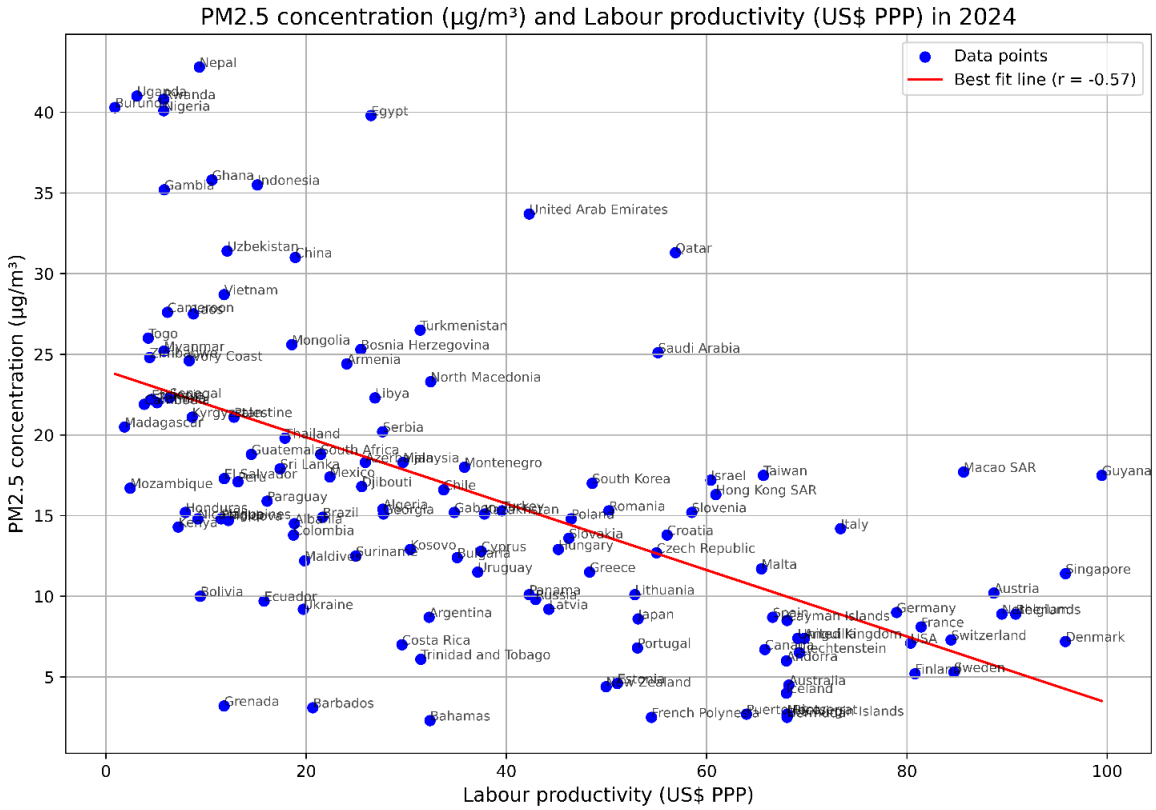


Figure 3: PM2.5 concentration (µg/m^3) and Labour productivity (US$ PPP) in 2024, data sources: IQAir [3] and ILO [21]

Year	2022
Correlation coefficient	0.457711479
R Squared	0.209499798
Intercept	24.46884366
Slope	-0.187742976
P value	4.97192E-07
Standard Error	9.30104086

Table 1: Related statistics for 2022

Year	2023
Correlation coefficient	0.531622601
R Squared	0.28262259
Intercept	24.02487391
Slope	-0.191959928
P value	6.94336E-10
Standard Error	7.960423087

Table 2: Related statistics for 2023

Year	2024
Correlation coefficient	0.573003939
R Squared	0.328333514
Intercept	23.96789092
Slope	-0.20571059
P value	2.36012E-12
Standard Error	7.887937008

Table 3: Related statistics for 2024

Data

Data used for the year 2024. PM2.5 values are in µg/m^3 and labour productivity in US$ PPP.

Year	2024	
Country	PM2.5	Labour productivity
Bahamas	2.3	32.36
French Polynesia	2.5	54.48
Bermuda	2.5	68.03
U.S. Virgin Islands	2.6	68.03
Montserrat	2.7	68.03
Puerto Rico	2.7	63.97
Barbados	3.1	20.66
Grenada	3.2	11.8

Iceland	4	67.96
New Zealand	4.4	49.95
Australia	4.5	68.2
Estonia	4.6	51.08
Finland	5.2	80.8
Sweden	5.3	84.7
Andorra	6	67.96
Trinidad and Tobago	6.1	31.44
Liechtenstein	6.5	69.26
Canada	6.7	65.81
Portugal	6.8	53.09
Costa Rica	7	29.57
USA	7.1	80.36
Denmark	7.2	95.83
Switzerland	7.3	84.41
Anguilla	7.4	69.75
United Kingdom	7.4	69.1
France	8.1	81.41
Cayman Islands	8.5	68.03
Japan	8.6	53.15
Argentina	8.7	32.27
Spain	8.7	66.58
Netherlands	8.9	89.48
Belgium	8.9	90.86
Germany	9	78.97
Ukraine	9.2	19.71
Latvia	9.2	44.24
Ecuador	9.7	15.79
Russia	9.8	42.91
Bolivia	10	9.42
Lithuania	10.1	52.83
Panama	10.1	42.27
Austria	10.2	88.68
Singapore	11.4	95.83
Greece	11.5	48.31
Uruguay	11.5	37.12
Malta	11.7	65.47
Maldives	12.2	19.84
Bulgaria	12.4	35.08
Suriname	12.5	24.95
Czech Republic	12.7	55
Cyprus	12.8	37.47
Hungary	12.9	45.18
Kosovo	12.9	30.4
Slovakia	13.6	46.24

Croatia	13.8	56.05
Colombia	13.8	18.73
Italy	14.2	73.37
Kenya	14.3	7.22
Albania	14.5	18.83
Moldova	14.7	12.22
Philippines	14.8	11.51
Poland	14.8	46.47
Nicaragua	14.8	9.19
Brazil	14.9	21.63
Kazakhstan	15.1	37.8
Georgia	15.1	27.71
Slovenia	15.2	58.5
Honduras	15.2	7.94
Gabon	15.2	34.79
Romania	15.3	50.23
Turkey	15.3	39.56
Algeria	15.4	27.66
Paraguay	15.9	16.1
Hong Kong SAR	16.3	60.91
Chile	16.6	33.73
Mozambique	16.7	2.41
Djibouti	16.8	25.54
South Korea	17	48.57
Peru	17.1	13.18
Israel	17.2	60.41
El Salvador	17.3	11.81
Mexico	17.4	22.37
Guyana	17.5	99.47
Taiwan	17.5	65.66
Macao SAR	17.7	85.65
Sri Lanka	17.9	17.39
Montenegro	18	35.81
Malaysia	18.3	29.67
Azerbaijan	18.3	25.91
Guatemala	18.8	14.52
South Africa	18.8	21.45
Thailand	19.8	17.87
Serbia	20.2	27.63
Madagascar	20.5	1.86
Kyrgyzstan	21.1	8.6
Palestine	21.1	12.79
Cambodia	21.9	3.82
Zambia	22	5.09
Ethiopia	22.2	4.54

Libya	22.3	26.87
Senegal	22.3	6.35
North Macedonia	23.3	32.43
Armenia	24.4	24.05
Ivory Coast	24.6	8.32
Zimbabwe	24.8	4.37
Saudi Arabia	25.1	55.14
Myanmar	25.2	5.79
Bosnia Herzegovina	25.3	25.45
Mongolia	25.6	18.56
Togo	26	4.22
Turkmenistan	26.5	31.38
Laos	27.5	8.74
Cameroon	27.6	6.14
Vietnam	28.7	11.8
China	31	18.91
Qatar	31.3	56.88
Uzbekistan	31.4	12.11
United Arab Emirates	33.7	42.27
Gambia	35.2	5.82
Indonesia	35.5	15.12
Ghana	35.8	10.59
Egypt	39.8	26.48
Nigeria	40.1	5.78
Burundi	40.3	0.9
Rwanda	40.8	5.78
Uganda	41	3.08
Nepal	42.8	9.34

Table 4: Data table, data sources: IQAir [3] and ILO [21]

Conclusion

The results show that the null hypothesis can be rejected with 95% confidence, suggesting there is a clear link between the concentration of PM2.5 and labour productivity. In general, countries that are more productive tend to have cleaner air. This could mean that healthier environments help people work better, or that successful countries invest more in keeping the air clean. Either way, it's a connection worth paying attention to.

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